READORA OPEN ACADEMIC TEXTBOOK SERIES

Thermodynamics & Heat Transfer

Classical Laws, Pure Substance Cycles & Thermal Power Systems

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Preface & Curriculum Overview

This textbook has been compiled for undergraduate and postgraduate students enrolled in the **Mechanical Engineering (MECH)** program. It provides structured, rigorous theoretical models, practical case demonstrations, analytical formulas, and chapter-end review exercises. Students are encouraged to practice the core derivations and apply algorithmic workflows to real-world engineering challenges.

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Chapter 1: The First and Second Laws of Thermodynamics

1.1 Conservation of Energy in Closed Systems

The First Law establishes energy as a conserved property. For any thermodynamic cycle, net heat transfer equals net work performed: delta U = Q - W.

> ***Key Concept:*** *Entropy is a state function that increases monotonically in isolated spontaneous natural processes.*

Review Questions & Laboratory Exercises

- Derive the Carnot thermal efficiency in terms of reservoir temperatures Th and Tc.
- Calculate the entropy change for an ideal gas undergoing isothermal expansion.

Chapter 2: Vapor & Gas Power Generation Cycles

2.1 The Ideal Rankine Steam Cycle

Rankine cycles form the foundation of thermal power plants comprising isentropic pumping, constant-pressure steam generation, expansion, and condensation.

> ***Key Concept:*** *Reheating and regenerative feed-water heating increase cycle thermal efficiency.*

Review Questions & Laboratory Exercises

- Draw the T-s and P-v diagrams for an ideal Brayton gas turbine cycle.
- Define the coefficient of performance (COP) for refrigeration cycles.

Chapter 3: Heat Transfer: Conduction, Convection & Radiation

3.1 Fourier Law of Thermal Conduction

One-dimensional steady heat conduction is governed by Fourier Law: q = -k * (dT/dx), where k is the material thermal conductivity.

> ***Key Concept:*** *Newton Law of Cooling governs convective surface heat flux proportional to film heat transfer coefficients.*

Review Questions & Laboratory Exercises

- Solve the steady-state radial heat conduction equation for cylindrical pipe insulation.
- State the Stefan-Boltzmann law for blackbody thermal radiation emission.